



Euclid's Theorem, Eureka! *Josh Hiller*

Said Euclid:

Let there be finitely many primes,
then we may produce a number greater than 1,
with no prime factor.
The fundamental theorem of arithmetic is broken.
The impossible must occur.
Eureka!

Said Furstenberg:

Let there be finitely many primes
then, in the evenly spaced integer topology
 $\{-1, 1\}$ is a set that is closed
and not closed.
The impossible must occur.
Eureka!

Said Meštrović:

Let there be finitely many primes,
then there are only two primes,
with 3 being the largest.
5, therefore, cannot be prime.
The impossible must occur.
Eureka!

Said Euler, said Erdős, said many more:

Let there be finitely many primes:
From the one statement
endless epiphanies.
Perhaps even one for every prime,
waiting to be uncovered,
so that time and time and time again
we may smile and declare:
Eureka!

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